CSCI 3060U – Winter 2025

Course Project Phase #6 – Integration and Delivery

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1. Overview

In this phase, we integrated a Front End system from a previous phase with our Back End to simulate a working day-to-day banking system. The goal was to process multiple transaction sessions, generate transaction files, and update account data over multiple days while keeping both systems independent.

2. System Design

The system is composed of two separate programs:

- **Front End (Phase 2)**
  Handles user interaction and generates transaction files (`.atf`).
- **Back End (Phase 5)**
  Processes the merged transaction file and updates the current accounts file.

Both programs are executed independently and communicate only through text files, as required.

Because the Front End and Back End use different file formats, we implemented conversion scripts to translate between the two formats.

3. Daily Script (`daily.sh`)

The Daily script simulates one day of banking operations.

Steps performed:

1. Runs the Front End multiple times using transaction session input files.
2. Each session produces its own Daily Transaction File (`.atf`).
3. All `.atf` files are concatenated into a single Merged Daily Transaction File.
4. The current accounts file is converted into the format required by the Back End.
5. The Back End is executed using the merged transaction file.
6. The updated accounts file is converted back into Front End format for use in the next day.

This ensures all communication between systems is done through text files.

4. Weekly Script (`weekly.sh`)

The Weekly script simulates seven days of system operation.

Steps performed:

1. Runs the Daily script seven times.
2. Each day uses a different set of transaction session files.
3. The output current accounts file from one day becomes the input for the next day.
4. This creates a continuous simulation of banking activity over a week.

5. File Flow

For each day:

1. **Input:** Current accounts file (Front End format)
2. Front End sessions generate multiple `.atf` files
3. `.atf` files are merged into one transaction file
4. File is converted to Back End format
5. Back End processes transactions
6. Updated accounts file is produced
7. File is converted back to Front End format for the next day

6. Design Decisions

- The Front End and Back End were kept completely separate to meet project requirements.
- Conversion scripts were used to handle differences in file formats instead of modifying previous phases.
- Session files were created to simulate realistic banking activity across multiple days.
- Error handling was added to ensure scripts do not silently fail when files are missing.

7. How to Run

Step 1: Compile Front End

javac -d Phase2/bin Phase2/src/*.java

Step 2: Run Weekly Simulation

bash Phase6/weekly.sh

Step 3: View Final Output

cat Phase6/output/day7/day7_currentaccounts_frontend.txt

Optional: View Transactions

cat Phase6/output/day1/day1_merged.atf

Running a Single Day (Daily Script)

You can also run just one day instead of the full week.

bash Phase6/daily.sh Phase6/sessions/day1 Phase6/data/frontend_currentaccounts.txt
Phase6/output/day1 day1 .

What this does

- Runs all session files in `sessions/day1`
- Generates a transaction file (`.atf`) for each session
- Merges them into one daily transaction file
- Runs the Back End using the merged file
- Produces an updated accounts file for the next day

View results

cat Phase6/output/day1/day1_merged.atf
cat Phase6/output/day1/day1_currentaccounts_frontend.txt